VadrevuDamini_HW8

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1.

(a)

- n= (10% of all matched pairs), which is the total number of discordant pairs,
- p=0.25, because in one-fourth of the discordant pairs, drug A leads to healing, while drug B does not.

Thus, the binomial random variable X follows a binomial distribution with parameters where n is 10% of all matched pairs) and p=0.25, symbolically represented as:

Y ~Binom(n,0.25).

(b)

```
library(EnvStats)
```

```
Warning: package 'EnvStats' was built under R version 4.3.3


Attaching package: 'EnvStats'

The following objects are masked from 'package:stats':

    predict, predict.lm
```

```
n_required <- propTestN(p.or.p1 = 0.025, p0.or.p2 = 0.075, power = 0.8,
                        sample.type = "one.sample", alternative = "two.sided",
                        alpha = 0.05)
```

```
Warning in propTestN(p.or.p1 = 0.025, p0.or.p2 = 0.075, power = 0.8, sample.type = "one.sample",
: The computed sample size 'n' is too small, relative to the given value of 'p', for the normal
approximation to work well for the following element indices:
     1
```

```
# Print the number of matched pairs required
print(n_required)
```

```
[1] 168
```

(c)

```r
power_200_pairs <- propTestPower(n.or.n1 = 200, p.or.p1 = 0.025, p0.or.p2 = 0.075,
                                 sample.type = "one.sample", alternative = "two.sided",
                                 alpha = 0.05, correct = FALSE)

# Print the power of the study with 200 matched pairs
print(power_200_pairs)
```

```
[1] 0.8892485
```

2

(a)

Let Xm represent the number of successes in the medical treatment group, where success is defined as ≤1 episode of OTM in the first 12 months after treatment. Then Xm follows a binomial distribution with parameters nm (number of observations in the medical group) andpM=0.50 (the assumed success rate in the medical group).

$$X_m \; Binom(n_m, 0.50)$$

Let XS represent the number of successes in the surgical treatment group. Then XS follows a binomial distribution with parameters nS (number of observations in the surgical group) and pS =0.70 (the assumed success rate in the surgical group).

$$X_s \; Binom(n_s, 0.70)$$

(b)

Null Hypothesis: Success rates are equal

Alternative hypothesis : Success rates are not equal.

```r
library(EnvStats)

# Given sample sizes for the medical and surgical groups
n_medical <- 100  # Medical group
n_surgical <- 50   # Surgical group

# Given success probabilities for the medical and surgical treatments
p_medical <- 0.5  # Medical treatment success probability
p_surgical <- 0.7  # Surgical treatment success probability

# Significance level and whether to apply continuity correction
alpha <- 0.05
continuity_correction <- FALSE
desired_power <- 0.9  # 90%
```

```
sample_sizes <- propTestN(p.or.p1 = p_surgical, p0.or.p2 = p_medical, sample.type = "two.sample",
                          ratio = 2)

n_surgical_calculated <- sample_sizes$n1
n_medical_calculated <- n_surgical_calculated * 2

# Output the required sample sizes
list(n_medical = n_medical_calculated, n_surgical = n_surgical_calculated)
```

```
$n_medical
[1] 188

$n_surgical
[1] 94
```

(c)

```
power <- propTestPower(n.or.n1 = n_surgical, n2 = n_medical, p.or.p1 = p_surgical,
                       p0.or.p2 = p_medical, sample.type = "two.sample",
                       alternative = "two.sided", alpha = alpha,
                       correct = continuity_correction)


power
```

```
[1] 0.6510791
```

3

(a)

Control Group:

$$Y_{control} \;\; Binom(n_{control}, 0.9985)$$

Study Group :

$$Y_{study} \, Binom(n_{study}, 0.9988)$$

(b)

```
propTestN(p.or.p1 = 0.9988, p0.or.p2 = 0.9985, sample.type = "two.sample",
          power = 0.80, alternative = "two.sided", correct = FALSE, ratio = 1)
```

```
[1] 235148
```

(c)

```
propTestN(p.or.p1 = 0.9940, p0.or.p2 = 0.9925, sample.type = "two.sample",
          power = 0.80, alternative = "two.sided", correct = FALSE, ratio = 1)
```

```
[1] 46775
```